

4. Limits

4.4. L'Hospital's Rule

4.4.3. Limit Examples (Continued)

4.4b Learning Objectives

- Understand how to transform any indeterminate form into one we can use l'Hospital's rule on
- More limit practice/reminders before next section

Evaluate the limits. Remember to list where and why you can use l'Hospital's rule if you do so.

Example 4.4.1

$$\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 + 2x}$$

Solution

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 + 2x} & \text{ has IFL infinity/infinity} \\ & \stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{2x}{3x^2 + 2} \quad (\text{still has IFL } \frac{\infty}{\infty}) \\ & \stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{2}{6x} \\ & = 0\end{aligned}$$

Example 4.4.2

$$\lim_{x \rightarrow 1^+} \frac{1}{\ln(x)} - \frac{1}{x-1}$$

Solution

$$\lim_{x \rightarrow 1^+} \frac{1}{\ln(x)} - \frac{1}{x-1} \text{ has IFL } \infty - \infty$$

We cannot do l'Hospital's rule without rearranging to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ first. Looking at the form of this, one thing we can try is adding (subtracting) the fractions together...Let's see if that helps us at all.

$$\begin{aligned}
\lim_{x \rightarrow 1^+} \frac{1}{\ln(x)} - \frac{1}{x-1} &= \lim_{x \rightarrow 1^+} \frac{1}{\ln(x)} \frac{x-1}{x-1} - \frac{1}{x-1} \frac{\ln(x)}{\ln(x)} \\
&= \lim_{x \rightarrow 1^+} \frac{x-1}{\ln(x)(x-1)} - \frac{\ln(x)}{\ln(x)(x-1)} \\
&= \lim_{x \rightarrow 1^+} \frac{x-1-\ln(x)}{\ln(x)(x-1)} \quad \text{has IFL } \frac{0}{0} \\
&\stackrel{\text{LH}}{=} \lim_{x \rightarrow 1^+} \frac{1-\frac{1}{x}}{\frac{1}{x}(x-1)+\ln(x)} \\
&= \lim_{x \rightarrow 1^+} \frac{1-\frac{1}{x}}{1-\frac{1}{x}+\ln(x)} \quad \text{still has IFL } \frac{0}{0} \\
&\stackrel{\text{LH}}{=} \lim_{x \rightarrow 1^+} \frac{\frac{1}{x^2}}{\frac{1}{x^2}+\frac{1}{x}} \\
&= \frac{1}{2}
\end{aligned}$$

Tip

Another tool we have to manipulate an expression from one IFL to another is to combine things into 1 fraction!

Note

Recall: If $f(x)$ is continuous and $\lim_{x \rightarrow a} g(x)$ exists, then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

Example 4.4.3

$$\lim_{x \rightarrow 0^+} (1 + \sin(4x))^{\cot(x)}$$

Solution

$$\lim_{x \rightarrow 0^+} (1 + \sin(4x))^{\cot(x)} \quad \text{has IFL } 1^\infty$$

This is not one of the indeterminate forms we can use l'Hospital's rule on. We don't really want ∞ in the exponent, the 2 forms we can do LH on are $\frac{0}{0}$ and $\frac{\infty}{\infty}$, and neither have exponents. What are some things that can take an exponent and move it somewhere else? That's right! Logarithms! We can't just take $\ln$ of something, so we use the fact that $e^{\ln(x)} = x$. Then,

$$\begin{aligned}
\lim_{x \rightarrow 0^+} (1 + \sin(4x))^{\cot(x)} &= \lim_{x \rightarrow 0^+} e^{\ln((1 + \sin(4x))^{\cot(x)})} \\
&= e^{\left(\lim_{x \rightarrow 0^+} \ln((1 + \sin(4x))^{\cot(x)})\right)}
\end{aligned}$$

This notation is really cumbersome though, so before we continue, we just look at the exponent. We have to come back and remember e at the end though.

Consider just the exponent now,

$$\lim_{x \rightarrow 0^+} \ln((1 + \sin(4x))^{\cot(x)}) = \lim_{x \rightarrow 0^+} \cot(x) \ln(1 + \sin(4x))$$

This now has indeterminate form of $\infty \cdot 0$ so we can try using $f(x) = \frac{1}{f(x)}$ again,

$$\begin{aligned} \lim_{x \rightarrow 0^+} \ln((1 + \sin(4x))^{\cot(x)}) &= \lim_{x \rightarrow 0^+} \cot(x) \ln(1 + \sin(4x)) \\ &= \lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin(4x))}{\tan(x)} \quad \text{has IFL } \frac{0}{0} \\ &\stackrel{\text{LH}}{=} \lim_{x \rightarrow 0^+} \frac{\frac{4 \cos(4x)}{1 + \sin(4x)}}{\sec^2(x)} \\ &= \frac{4}{1} = 4 \end{aligned}$$

But wait! This is not our final answer! Recall

$$\begin{aligned} \lim_{x \rightarrow 0^+} (1 + \sin(4x))^{\cot(x)} &= \lim_{x \rightarrow 0^+} e^{\ln((1 + \sin(4x))^{\cot(x)})} \\ &= e^4 \end{aligned}$$

Exercise 4.4.1

Show

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

4.4b Section Summary:

- We computed more limits with other indeterminate forms besides $\frac{0}{0}$ and $\frac{\infty}{\infty}$